Jingyun Yang

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Machine Learning · GPA: 4.0 — *August 2020 – December 2021*

University of Southern California — Los Angeles, CA
Bachelor of Science in Computer Science · Minor in Mathematics · GPA: 4.0 — *August 2016 – May 2020*

Publications

1. Hsiao-Yu Fish Tung*, **Jingyun Yang***, Yunchu Zhang*, Gaurav Pathak, Ashwini Pokle, Christopher G. Atkeson, Katerina Fragkiadaki, "Visually-Grounded Library of Behaviours for Generalizing Manipulation Across Objects, Configurations, and Views", In submission.
2. Youngwoon Lee, **Jingyun Yang**, Joseph J. Lim, "Learning to Coordinate Manipulation Skills via Skill Behavior Diversification", International Conference on Learning Representations (ICLR), 2020.
3. Karl Pertsch, Oleh Rybkin, **Jingyun Yang**, Konstantinos Derpanis, Kostas Daniilidis, Joseph J. Lim, Andrew Jaegle, "KeyIn: Keyframing for Visual Planning", Learning for Dynamics and Control (L4DC), 2020.

Experience

CMU Professor Katerina Fragkiadaki's Lab — Pittsburgh, PA
Research Assistant · Supervisor: Katerina Fragkiadaki — *June 2020 – Present*

- **Visual Learning for Manipulation:** Investigating frameworks that learn appropriate visual representations to help improve sample efficiency and generalization in robot manipulation tasks.

USC Cognitive Learning for Vision and Robotics (CLVR) Lab — Los Angeles, CA
Research Assistant · Supervisor: Joseph J. Lim — *August 2018 – May 2020*

- **Compositional Imitation Learning:** Researched on latent space models that learn useful robot skills from a large set of demonstrations to achieve efficient hierarchical learning of a downstream task.
- **Learning Coordinated Tasks from Rewards:** Designed, implemented, and tested a hierarchical reinforcement learning model that tackles complex tasks of multiple agents by learning composable primitive skills and coordination of the skills. Paper accepted in ICLR 2020.
- **Video Prediction with Temporal Hierarchy:** Developed experiments and designed downstream planning algorithm for a model that learns to discover informative frames in a future video sequence and represent the video via its keyframes.

USC Integrated Media Systems Center / USC Spatial Sciences Institute — Los Angeles, CA
Research Assistant · Supervisor: Yao-Yi Chiang — *August 2016 – August 2018*

- **Neural Style Transfer:** Experimented with deep learning methods that transform a map dataset with arbitrary styles to a dataset with target styles required for text detection training.
- **Text Detection:** Experimented with various state-of-the-art text detection models on historical map datasets.
- **Timeseries Data Analytics:** Analyzed dynamic and large-scale transportation sensor data sent by buses in LA.

Bloomberg L.P. — New York, NY
Software Engineering Intern · Supervisor: Luke Shen — *May 2018 – August 2018*

- **Backend Software Development:** Migrated core functionalities of the Company Financials Core Algorithms Library to Spark in Scala.

USC Information Sciences Institute — Los Angeles, CA
Research Assistant · Supervisor: Yolanda Gil — *May 2017 – August 2017*

- **Data Visualization:** Created a module that automatically generates visualizations for either an arbitrary dataset or a group of datasets in the workflow system.

Awards and Honors

- USC University Trustees Award — *April 2020*
- USC Provost's Undergraduate Research Fellowship — *May, August 2019*
- Tau Beta Pi the Engineering Honor Society — *April 2018*